STIMULATIONS VS. PRACTICAL

PROS AND CONS - STIMULATION LED AND PROTOTYPE LED:

❖ STIMULATION LED:

Simulation-led design uses computer software and mathematical models to predict aircraft behavior before manufacturing the physical aircraft.

HOW IT WORKS: The designer enters parameters such as airfoil shape, wing dimensions, mass estimates, speed, and operating conditions. Software estimates aerodynamic forces, stability, structural behavior, or other performance characteristics.

Advantages

1. **Reduces material waste**: Several designs can be compared before building them.
2. **Allows early optimization**: Wing area, aspect ratio, airfoil, and tail dimensions can be adjusted digitally.
3. **Provides useful predictions**: Lift, drag, pressure distribution, and stability trends can be studied.
4. **Improves design understanding**: Engineers can investigate why a design may perform poorly.
5. **Makes comparisons easier:** Multiple design options can be evaluated using consistent assumptions.

Limitations

1. Results depend on the accuracy of the input data and mathematical models.
2. Simplified models may not capture complex airflow, turbulence, or separation accurately.
3. Actual construction may introduce misalignment, surface roughness, excess mass, and structural flexibility.
4. Some advanced software requires powerful computers, expertise, or expensive licences.
5. A simulation cannot fully reproduce real-world take-off, control response, radio problems, or manufacturing defects.

❖ PROTOTYPE LED:

Prototype-led design involves manufacturing an actual aircraft or test component early, testing it, and improving it based on observed performance.

HOW IT WORKS: The team builds a prototype using materials such as foam board, balsa wood, plywood, or composite materials. It then measures mass, checks geometry, tests the propulsion system, and performs controlled flight tests.

Advantages

1. Reveals real construction and assembly problems.
2. Measures actual aircraft mass, motor thrust, and control-system response.
3. Detects issues that simplified simulations may miss.
4. Provides direct evidence of take-off, handling, and flight performance.
5. Helps the team develop practical manufacturing and troubleshooting skills.

Limitations

1. Materials and components may be wasted when a design fails.
2. Manufacturing and testing take physical effort and workshop time.
3. Poorly controlled tests may produce misleading results.
4. Rebuilding an entire aircraft for each change can be slow.
5. Flight tests require suitable weather, space, supervision, and safety precautions.

CONCLUSION: Neither approach is sufficient alone. Simulation is most useful for comparing alternatives and reducing avoidable mistakes; practical prototyping is essential for verifying the actual aircraft.

Comparison Between Stimulations vs. Practical Led:

The table given below compares some important features of aircraft building and constraints in its building:

Factor	Simulation-led design	Prototype-led design
Cost	Usually lower for comparing many designs; software may require training or paid licences.	Higher material and component costs if repeated builds are needed.
Time	Fast for changing dimensions and comparing concepts, once the model is ready.	Slower when parts must be fabricated, assembled, and tested.
Accuracy	Predictive; depends on model assumptions, input data, and method.	Measures actual behavior, but instrument errors and test conditions matter.
Iteration speed	High for digital changes and parameter sweeps.	Moderate or low if structural changes require rebuilding.
Failure detection	Can identify predicted aerodynamic, stability, and structural issues.	Can reveal real assembly, propulsion, control, and operational problems.
Final performance	Helps optimize a theoretical design.	Confirms the performance achieved by the manufactured aircraft.

NOTE: A SIMPLE PHYSICAL TEST CAN BE QUICKER THAN A COMPLICATED SIMULATION, AND A CAREFULLY VALIDATED SIMULATION CAN BE MORE USEFUL THAN AN UNCONTROLLED PHYSICAL TEST.

I recommend **a simulation-first, prototype-verified hybrid** approach.

- Use simulations to eliminate poor design options before spending money.
- Build an early prototype as soon as the basic design meets the requirements.
- Test the prototype and compare measured results with predictions.
- Correct the most important discrepancies before producing the competition-ready aircraft.

This approach balances cost, speed, accuracy, and practical reliability.

THE WORKFLOW OF THE PROJECT OF RC PLANE AND HOW IS PERFORMED:



NOTE: THE REFINING AND DESIGNING STEP CAN BE REPEATEDLY PERFORMED IF REQUIRED.

Let us discuss the preliminary calculations that are to be performed:

- Estimate the total mass, wing area, wing loading, stall speed, and required thrust. These calculations help reject unsuitable concepts before detailed design.
- For example; **WING LOADING** is,

$$\textbf{Wing loading} = \frac{W}{S}$$

W = **aircraft weight (newtons)** S = **wing surface area**

- The stall speed in steady level flight is calculated as;

$$V_s = \sqrt{\frac{2W}{\rho S C_{L_{\max}}}}$$

W =**aircraft weight(newtons)** S =**wing surface area**

ρ =**density of air** $C_{L_{\max}}$ =**maximum lift coefficient**

- The aspect ratio it affects the induced drag and wing geometry:

Aspect ratio= (span of wing)^2/ (area of the wing)

- These are some of the important values to be calculated. Other than this also there are various values which are important like **lift force, drag force, coefficient of lift, coefficient of drag, lift to drag ratio** and many more important parameters exist.

For **designing** we can use softwares like **Autocad, Solidworks**.

For **Running Stimulations** one can use **XFOIL, Ansys, Matlab & Simulink, FlexSim**.

Building the **actual Prototype** includes countering towards **measurement of actual component masses** and **inspection of warping, asymmetry, loose connections, and dimensional errors**.

Conduct physical tests: Verify the **centre of gravity** against the approved design range, **control- surface direction and travel**, **structural integrity, radio operation, and propulsion performance**.

Conduct controlled flight tests. Only proceed after the aircraft passes its ground checks. Begin with a suitable low-risk test plan and an experienced, authorized operator. Record **take-off behavior**, **climb, control response, and stability.**

Refine the design. Identify the largest difference between **prediction and measurement**. To resolve the issues start by **changing one suspected variable at one time** and while implementing a new one also first check its stimulation and then fix it.

PARAMETERS CHOOSEN FOR STIMULATION CHECK AND ITS VERIFICATION THROUGH PHYSICAL PROTOTYPE:

The team should choose parameters that directly influence the competition mission rather than spending time analysing every possible variable.

❖ Parameters to study through simulation

Parameter	What to investigate	Why it matters
1. Lift coefficient (C_L)	Lift across the expected operating angles of attack and speeds	Determines whether the wing can generate sufficient lift.
2. Drag coefficient (C_D)	Aerodynamic drag at expected flight conditions	Helps estimate the power and thrust needed to overcome drag.
3. Stall characteristics	Predicted maximum lift and changes near stall	Helps assess low-speed handling, while recognizing model limitations.
4. Static stability	Pitching moment, neutral point and centre-of-gravity sensitivity	Helps determine whether the aircraft tends to recover from small disturbances.
5. Structural strength	Estimated stress and deflection under relevant loads	Helps identify potential structural weaknesses.
6. Mass properties	Predicted mass distribution and centre of gravity	Affects balance, stability, and control response.

❖ Parameters to verify physically

Parameter	How to verify it	What to compare
1. Actual aircraft mass	Weigh the completed aircraft	Measured mass vs mass budget
2. Centre of gravity (CG)	Balance the aircraft at the specified measurement points	Actual CG vs approved design range
3. Motor thrust	Use a suitable, securely mounted thrust-measurement setup	Measured thrust vs propulsion estimate
4. Wing geometry	Measure span, chord, incidence and alignment	Built geometry vs CAD model
5. Control-surface movement	Check direction, travel and servo response	Actual movement vs control settings
6. Flight performance	Record flight speed where measurable, climb behavior, and mission results	Actual behavior vs predicted performance

Explain how simulations should be validated

Simulation validation means checking whether a model's predictions agree sufficiently with measurements from the real world.
The process should be systematic:

Step 1 — Establish a reference design.

Create the exact aircraft geometry, airfoil, estimated mass, CG, control settings, motor and propeller configuration used in the simulation.

Step 2 — Record predicted values.

Save the simulation outputs before physical testing. These might include estimated stall speed, lift coefficient, drag coefficient, pitching moment, and required thrust.

Step 3 — Measure the prototype.

Use calibrated instruments where possible. Measure mass using a scale, CG using a balancing method, and static thrust using a suitable thrust stand. Record the conditions under which each measurement was made.

Step 4 — Match the conditions.

Compare like with like. Aerodynamic results depend on factors such as airspeed, air density, angle of attack, Reynolds number, and configuration. Differences in test conditions can explain apparent disagreement.

Step 5 — Calculate the discrepancy.

For a predicted value $\mathbf{P}$ and measured value $\mathbf{M}$, calculate the percentage difference as:

$$\textbf{Difference (\%)} = \frac{|M - P|}{|P|} \times 100$$

NOTE: THIS FORMULA IS APPROPRIATE WHEN THE PREDICTED VALUE IS NONZERO AND IS A MEANINGFUL REFERENCE. FOR QUANTITIES NEAR ZERO, USE AN ABSOLUTE DIFFERENCE OR A SUITABLE UNCERTAINTY-BASED COMPARISON INSTEAD.

Step 6 — Investigate discrepancies.

Possible explanations include inaccurate input data, simplified models, measurement uncertainty, manufacturing differences, and environmental conditions.

The team should investigate the installed propeller, battery voltage under load, motor performance, and test setup rather than assuming the simulation is correct.

Step 7 — Update and retest.

Correct the inputs or model assumptions when evidence supports doing so. Repeat measurements if needed, and rerun the simulation using the revised information.

NOTE: THE MEASURED VALUES CAN ALSO CONTAIN SOME ERROR DUE THE INSTRUMENTS LEAST COUNT AND OTHER LIMITATIONS.

CASE STUDY: The Stimulated Plane Performs Well But The Prototype Cannot Take-Off – Plausible Causes And A Systematic Diagnostic Plan.

A. Plausible causes:

1.Incorrect mass or centre of gravity

The completed aircraft may be heavier than the model, or its CG may lie outside the approved design range. This changes the lift required and can make control difficult.

2. Insufficient installed thrust

The real propulsion system may produce less thrust than expected because of battery voltage drop, an unsuitable propeller, motor problems, or excessive installation losses.

3. Incorrect wing geometry

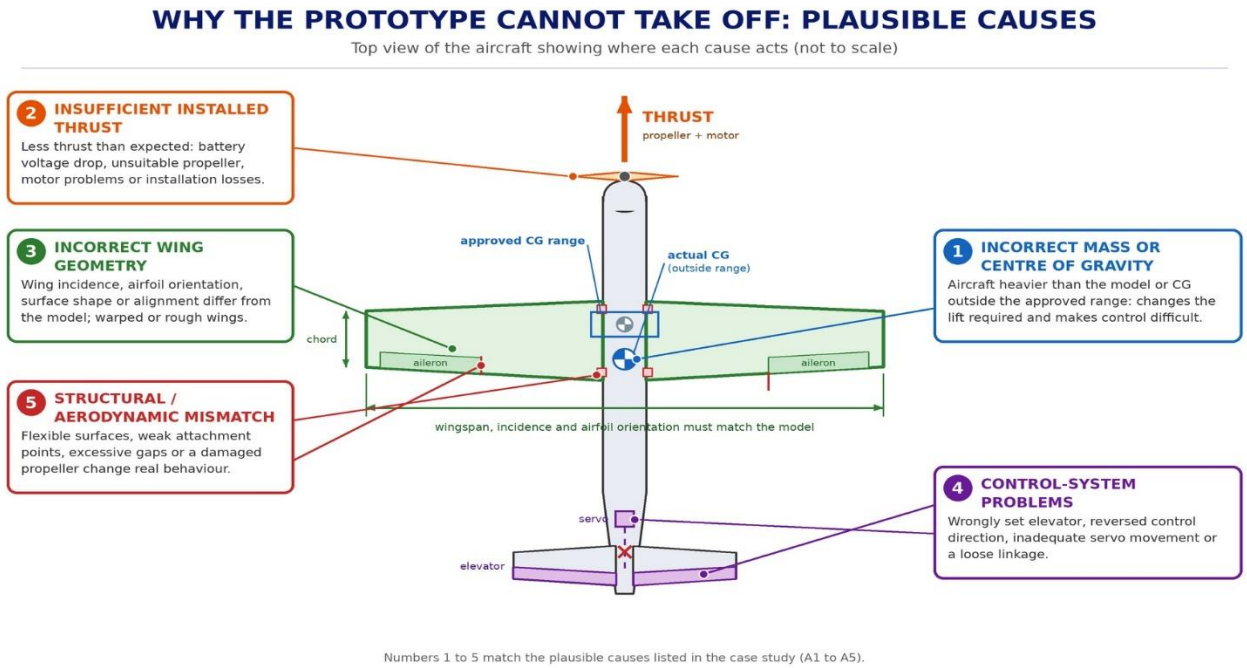
Wing incidence, airfoil orientation, surface shape, or alignment may differ from the simulated model. Warped wings or rough construction may also affect performance.

4. Control-system problems

An incorrectly configured elevator, a reversed control direction, inadequate servo movement, or a loose linkage may prevent the aircraft from rotating or responding properly.

5. Structural or aerodynamic mismatch

Flexible surfaces, weak attachment points, excessive gaps, or a damaged propeller can cause the real aircraft to behave differently from the model.



B. Systematic diagnostic plan

Check 1: Confirm the aircraft's actual condition

Weigh the aircraft, measure the CG, inspect the wing and tail alignment, and compare the finished aircraft with the design drawings.

Check 2: Verify propulsion

Inspect the motor, propeller, battery, wiring, and speed controller. Measure static thrust with a suitable test fixture and compare it with the estimate. Keep clear of the propeller and follow appropriate supervision and safety procedures.

Check 3: Verify controls

With propulsion made safe, check transmitter response, servo direction, control-surface travel, linkage security, and radio reliability.

Check 4: Recalculate take-off requirements

Use the measured aircraft mass and realistic estimates of lift and installed thrust. Check whether the predicted take-off speed and available ground distance are consistent with the actual setup and conditions.

Check 5: Compare the simulation with reality

Verify that the simulation uses the actual wing dimensions, mass, CG, airfoil, control configuration, and propulsion assumptions. Correct any inaccurate inputs.

Check 6: Retest before returning to flight

Correct the identified fault, repeat the relevant ground checks, and obtain approval from the responsible experienced operator before conducting another controlled flight test.

C. Decision logic: The approach towards solving the issue.

- If the aircraft is too **heavy**, investigate mass reduction without compromising structural integrity.
- If measured **thrust is too low**, identify the propulsion-system cause before redesigning the wing.
- If **CG or wing alignment is incorrect**, correct the construction or mass distribution and recheck stability.
- If **controls are faulty**, fix the control system before any flight test.
- If all physical checks pass but performance remains poor, revisit the **aerodynamic model and test assumptions.**

SOFTWARES FOR DESIGN AND STIMULATIONS

There exists various softwares that help designing of model and running stimulations at various conditions and parameters Some of them are described below with there main purpose of use and how they contribute to a project of building an RC Plane.

Icon	Software	Main purpose	How it helps in the project
	**XFLR5**	Airfoil and wing aerodynamic analysis	Estimates lift, drag, lift-to-drag ratio, and stability characteristics for suitable flight conditions.
	**OpenVSP**	Aircraft geometry modelling and aerodynamic analysis	Helps create a 3D aircraft model and evaluate aerodynamic characteristics.
	**Fusion 360**	CAD design and mechanical modelling	Designs the fuselage, wing structures, motor mounts, and other components.
	**SolidWorks**	3D CAD and structural design	Models aircraft components and checks assemblies, loads, and structural strength using available simulation features.
	**ANSYS**	Advanced engineering simulation	Can analyse structural stresses, deformation, and airflow using appropriate structural and computational fluid dynamics (CFD) tools.

Realistic Four Week Development Schedule:

We can divide above mentioned work flow in a 4 week schedule.

WEEK 1: Understanding the mission definition, performing all preliminary calculations and preparing a design of aircraft using various softwares.

- Study competition rules and set measurable requirements.
- Estimate mass, wing loading, wing area, stall speed, and propulsion needs; that is all the required calculated values.
- Choose candidate airfoils and basic wing configurations whose features suits our mission the best.
- Prepare CAD geometry and an initial mass budget.

Milestone: Preliminary designs approved.

WEEK 2: Running of stimulations and preparing a prototype of the aircraft.

- Run aerodynamic and preliminary stability analyses.
- Finalize the wing, tail, and structural layout.
- Procure and check the required components.
- Manufacture the wing, fuselage, and tail; begin assembly.

Milestone: Initial prototype assembled

WEEK 3: Ground testing and initial flight testing

- Complete assembly and measure mass and CG.
- Verify propulsion, control surfaces, radio operation, and structure.
- Conduct approved controlled flight tests if the aircraft passes all checks and conditions permit.
- Compare measured results with simulation predictions.

Milestone: First test results recorded

WEEK 4: Refinement, contingency, and final checks

- Reserve the first part of the week for repairs and unresolved issues.
- Correct significant differences between measured and predicted performance.
- Repeat relevant ground and flight tests.
- Freeze the final configuration, document the results, and prepare for the competition.

Milestone: Verified competition-ready aircraft

- ❖ A REALISTIC SCHEDULE MUST ALLOW FOR SETBACKS RATHER THAN ASSUMING EVERY TASK SUCCEEDS ON THE FIRST ATTEMPT.

Possible setback	Planned response
Component delivery delays	Identify alternatives compatible with the design and confirm availability early.
Excessive aircraft mass	Recheck the mass budget and remove unnecessary components where safe.
Prototype assembly errors	Keep time available in Weeks 3 and 4 for correction and reinspection.
Poor take-off or flight performance	Follow the diagnostic plan in Step 7; do not skip safety checks to save time.
Unfavourable weather	Use the time for analysis, documentation, ground checks, and repairs; reschedule flight tests.